gbsv Mini-Challenge 2

2026-04-18

General Information

In this mini-challenge, the tasks are in the form of experiments in foundations in signal and image processing to assess LO1-LO3 themed to a country of choice. Each student submits a unique solution with individual data and analysis. That is, each student chooses signals or images themed for the selected country and the assigned tasks. Unless otherwise mentioned, existing libraries may be used.

Note: Some tasks are kept open. Set your own frame for the limitation or ask if the boundaries are not clear to you. As a rule of thumb one day should take you around 2h +/-30min per day.

Version Individual Rhythm

- Conduct your MC according to your own schedule/rhythm
- Mandatory check-in (review meeting): MC is only graded if a check-in was conducted.

Version 15-Day Challenge

- Profit from regular feedback
- Earn 1 bonus credit per submitted day if you submit at least 10 days
- Bonus credits: up to +0.33 added to final grade (scaled to 15; capped at 6.0 for maximum grade)
- Revisions are allowed until day 15 if changes are clearly highlighted in color. Code does not need to be colored, but a colored text tag should mark start and end of code changes.
- Submission deadline: all entries must be submitted by day 15 (per challenge)
- Have a look at what order of “competences” / days makes most sense to you. Some prefer to start with methods rather than data and domain. Feel free to revert that order.

Submission and Grading

- Submission are conducted via Microsoft Teams Assignments in the gbsv Teams.
- Deliverables:
    - Submit notebook source code and HTML if output beneficial (one notebook per subtask)
    - Link to code and results incl. signals/images, if applicable
- Make sure that audios are playable within notebook exports
- See grading rubrics, gbsv interpretations and indicators for strong vs. weak performance
- Submissions received after the deadline, zipped or submitted via email will not be considered.
- Assessment is based on the quality of your technical framing and justification, not on whether you adopt a provided example use case or develop your own.
- ➢ Effective selection of task-appropriate data (core skill), pinpoint creativity, moderate diversity, and critical thinking are sought rather than extensive submissions.

1.1 Augmentation

Learning Objectives

- Methodological: Select and justify suitable augmentation procedures and parameter settings based on image characteristics and the defined problem.
- Technical: Implement augmentation methods and apply them in a structured and traceable manner.
- Analytical: Quantitatively assess the effect of augmentations using an appropriate metric and compare configurations.
- Reflective: Evaluate the usefulness, limitations, and risks of augmentation in relation to the defined use case.

Day 1 – Data & Domain

Define the application context in which image augmentation is relevant. Formulate a clear use case and problem statement related to your selected country. Describe a realistic challenge in this context that motivates the need for augmentation (e.g., limited data variability, sensitivity to noise or lighting conditions, domain shift).

Then select one representative 2D image for this use case. Describe:

- its origin and content
- the relevant visual structures (e.g., edges, textures, objects)
- Identify one clearly identifiable structure or region in the image (e.g., an object, boundary, or texture) and explain how the previously defined challenge manifests specifically in this structure.

Finally, define the objective of augmentation in this context (e.g., simulate realistic distortions, increase variability, improve robustness).

No augmentation methods should be introduced at this stage.

Start a new notebook.

Reference examples: The following use cases are provided for orientation. You may select one of them and adapt it to your chosen country, or define a suitable alternative within your selected domain.

Example 1: Urban Traffic Monitoring – License Plate Visibility

- **Use Case:** Traffic cameras in a major city are used to automatically detect and read license plates for traffic analysis and enforcement.
- **Problem Statement:** Images are often affected by varying lighting conditions (e.g., night scenes, glare, shadows) and sensor noise, which can reduce the visibility of relevant features. Models trained on clean data may fail under these conditions. Augmentation is needed to simulate realistic degradations and improve robustness.

Example 2: Historical Document Digitization – Text Legibility

- **Use Case:** Scanned historical documents are processed to extract textual information for archival purposes.

- **Problem Statement:** Old documents often suffer from fading, stains, and uneven illumination. These artifacts reduce contrast and obscure textual content. Augmentation can simulate such degradations to improve robustness of downstream processing methods.

Day 2 – Methodological Design

Define and justify your augmentation strategy in direct relation to your problem statement, selected image, and identified structure. Select 1-2 augmentation procedures (excluding geometric transformations such as translation, rotation, or mirroring). The augmentations may operate in the spatial or frequency domain. For each augmentation:

- explain the underlying transformation principle
- justify why it is appropriate for the specific challenge and identifiable structure defined in Day 1
- define 1-2 key parameters and explain their expected effect on the image and on the identified structure

Clearly describe what type of change you aim to induce with each augmentation (e.g., noise characteristics, contrast variation, texture alteration), and how this relates to your objective. Based on these choices, define a small number of clearly structured configurations (e.g., baseline + 1-2 variations) by specifying concrete parameter settings that enable controlled comparison.

The focus of this day is on method selection, parameter reasoning, and structured experimental design. No implementation, visual results, or evaluation methods should be included.

Day 3 – Technical Implementation

Implement and apply your augmentation procedures as defined in Day 2. First, implement your selected augmentations (libraries may be used) and verify their correct functionality on a simple test image. Then apply the augmentations to your selected image using the predefined configurations from Day 2. Ensure that:

- all parameter settings are explicitly stated and traceable
- the sequence of transformation steps can be clearly followed
- intermediate and final outputs are clearly visualized

Focus on a small number of clearly structured configurations. Avoid generating many loosely connected variants.

The focus of this day is on correct, transparent, and traceable implementation. No quantitative evaluation or interpretation should be included.

Day 4 – Evaluation

Quantitatively evaluate the effect of your augmentations. Select exactly one quantitative evaluation measure to compare the original and augmented image. Clearly:

- define what the measure computes
- justify why it is appropriate for assessing the effect of your augmentations in relation to your objective and identified structure

Apply this measure consistently across all predefined configurations from Day 2.

Report:

- the metric values for each configuration
- a clear comparison relative to a baseline configuration

Ensure that:
- the same measure is used consistently
- results are presented in a clear and comparable form
- differences between configurations are explicitly quantified

Focus on a small number of configurations and a single, well-justified metric rather than multiple loosely connected measures.

The focus of this day is on objective, quantitative assessment and structured comparison.
No new simulations, theoretical derivations, or interpretative discussion should be introduced.

Day 5 – Analytical & Communication

On this day, you will observe, interpret, and discuss your findings in direct relation to your initial problem statement and defined use case. Based on the quantitative evaluation from Day 4, analyze how the applied augmentations affect the image and how these effects translate into domain-specific consequences. Structure your analysis into:
- **Observations**: Describe the quantitative results only. Refer explicitly to the metric values and how they change across configurations.
- **Interpretation**: Explain why the observed changes occur based on the applied methods and parameter choices. Relate these effects to relevant image characteristics in your use case.
- **Discussion and Critical Reflection**: Assess the practical applicability of your results. Discuss which configurations are suitable for the use case and why, how parameter choices influence this, and what limitations or risks arise when the augmentation is applied inappropriately. Reflect on the extent to which your evaluation measure captures the objective defined in Day 1.

Present your analysis in a concise, technically precise, and logically structured form (approximately 150-200 words). All arguments must clearly connect back to the original problem statement and use case. No new simulations or computations should be introduced.

1.2 Algorithms for Pattern Detection

Learning Objectives

- Conceptual: Apply classical image processing algorithms to detect patterns such as edges, lines, circles, or corners in a domain-specific context.
- Methodological: Select and justify an appropriate pattern detection algorithm and parameter configuration based on the pattern of interest and image characteristics.
- Technical: Implement the algorithm step-by-step and demonstrate relevant intermediate processing results in a structured and traceable manner.
- Analytical: Quantitatively assess detection performance using an appropriate measure and compare configurations.
- Reflective: Evaluate the effectiveness, limitations, and practical applicability of the algorithm for the defined use case, including sensitivity to parameters and image conditions.

Day 6 – Technical Implementation

Select one classical image processing algorithm for detecting patterns such as edges, lines, circles, or corners (e.g., Sobel, Canny, Hough Transform, Harris Corner Detector). The selected algorithm should be suitable for detecting a pattern with clear relevance in a real-world image analysis context related to your chosen country.

Implement the algorithm, ensuring that all relevant processing steps are explicitly represented. The individual steps may be self-programmed or implemented using libraries, but the sequence of operations must be clearly identifiable and traceable. Apply your implementation to a simple test image to verify correct functionality. Ensure that:

- the individual steps of the algorithm are clearly separated
- intermediate results of each step are explicitly visualized (e.g., gradients, thresholded images, accumulator space)
- all parameters are clearly defined and traceable

The objective of this day is to understand and demonstrate the structure of the algorithm, not to optimize performance or apply it to a fully defined use case. No problem statement, detailed use case, or quantitative evaluation should be introduced at this stage.

Start a new notebook.

Day 7 – Data & Domain

Define the application context for your implemented pattern detection algorithm. Formulate a clear use case and problem statement related to your chosen country. The use case must be consistent with the pattern type targeted by your algorithm (e.g., edges, lines, circles, corners) and reflect a realistic scenario in which detecting such patterns is meaningful. Select one representative 2D image and describe:

- its origin and content
- the relevant visual characteristics (e.g., edges, geometric structures, textures)

Clearly identify the pattern of interest in the image and explain how it appears in this specific context.

Explain why detecting this pattern is important for your defined use case and how the problem manifests in the selected image. Define the objective of pattern detection in this context (e.g., boundary

extraction, localization of structures, feature identification). The use case, image, and pattern must be coherently aligned with the algorithm implemented in Day 6.

No parameter tuning, evaluation, or interpretation of results should be included at this stage.

Reference examples: The following use cases are provided for orientation. You may select one of them and adapt it to your chosen country, or define a suitable alternative within your selected domain.

Example 1: Infrastructure Monitoring – Circular Structure Detection (Manholes / Road Signs)

- **Use Case:** Images captured in urban environments are used to monitor road infrastructure and support maintenance planning. Street-level images or inspection footage are analyzed to detect circular structures such as manhole covers, traffic signs, or round inspection lids.
- **Problem Statement:** Urban images often contain circular patterns corresponding to infrastructure elements. These structures may be partially occluded, affected by shadows, or embedded in complex backgrounds, making reliable detection challenging. Choose one:
    - Edge Detection (e.g., Sobel, Canny): The goal is to identify intensity transitions that outline potential circular structures. Detected edges provide the necessary input for identifying geometric shapes.
    - Circle Detection (e.g., Hough Transform): The objective is to detect circular patterns by accumulating evidence for circle parameters (center and radius) based on edge information. Detected circles correspond to physically meaningful infrastructure elements.
- The task is to analyze how well circular structures can be detected under varying image conditions and to interpret the results in relation to infrastructure monitoring.

Example 2: Cultural Heritage Analysis – Edge, Line, and Corner Detection in Artworks

- **Use Case:** Digital images of cultural heritage artifacts such as paintings or architectural details are analyzed to study visual structure and support documentation or restoration. Images from museums or public archives are used to identify structural elements such as contours, geometric patterns, and compositional features.
- **Problem Statement:** Artworks and cultural artifacts contain structural patterns such as edges, lines, and corners that define shapes, textures, and geometric elements. These patterns may be affected by aging, lighting conditions, or surface degradation, making reliable detection challenging. Choose one:
    - Edge Detection (e.g., Sobel, Canny): The goal is to detect intensity transitions corresponding to object boundaries, contours, or brush stroke structures. These edges represent fundamental visual features of the artwork.
    - Line Detection (e.g., Hough Transform): The objective is to detect linear structures such as architectural elements, perspective lines, or repeated geometric patterns. Detected lines correspond to compositional features within the artwork.
    - Corner Detection (e.g., Harris): The goal is to detect distinctive points such as intersections of lines or geometric features. These corners represent localized structural elements relevant for analyzing composition or structure.
- The task is to analyze how well structural elements of cultural artifacts can be detected and to interpret the results in relation to cultural heritage analysis and documentation.

Day 8 – Methodological Design

Define and justify your methodological approach for applying the pattern detection algorithm to your selected image. Based on your implemented algorithm and defined use case, specify how the algorithm will be applied to detect the pattern of interest. Identify 1-3 key parameters of your algorithm (e.g., thresholds, kernel size, accumulator resolution) and explain their expected effect on the detection outcome in relation to your image characteristics and pattern of interest.

Based on these choices, define a small number of clearly structured configurations (e.g., baseline + 1-2 variations) that enable controlled comparison. Avoid generating many loosely connected parameter variations. Justify why these parameter choices and configurations are appropriate for your use case and image.

The focus of this day is on methodological reasoning and structured experimental design.
No implementation, results, or evaluation should be included.

Day 9 – Evaluation

Quantitatively evaluate the performance of your pattern detection algorithm. Select exactly one quantitative evaluation measure that is appropriate for assessing the detection of your chosen pattern (e.g., number of detected features, detection accuracy relative to a reference, edge density, or another justified measure). Clearly:

- define what the measure computes
- justify why it is suitable for evaluating detection performance in relation to your pattern of interest and use case

Apply this measure consistently across all predefined configurations from Day 8.

Report:

- the metric values for each configuration
- a clear comparison relative to a baseline configuration

Ensure that:

- the same measure is used consistently
- results are presented in a clear and comparable form
- differences between configurations are explicitly quantified

Limit your analysis to a small number of clearly structured configurations.

The focus of this day is on objective, quantitative assessment and structured comparison.
No new simulations, theoretical derivations, or interpretative discussion should be introduced.

Day 10 – Analytical & Communication

On this day, you will observe, interpret, and discuss your findings in direct relation to your initial problem statement and defined use case. Based on the quantitative evaluation from Day 9, analyze how well your algorithm detects the intended patterns and how these results translate into domain-specific consequences. Structure your analysis into:

- **Observations**: Describe the quantitative results only. Refer explicitly to the metric values and how they change across configurations.

- **Interpretation**: Explain why the observed changes occur based on the algorithm steps and parameter choices. Relate these effects to relevant image characteristics in your use case.
- **Discussion and Critical Reflection**: Assess the practical applicability of your results. Discuss which configurations are suitable for the use case and why, for which elements or regions the algorithm works well or poorly, and how parameter choices influence this. Reflect on limitations, risks, and the extent to which your evaluation measure captures detection performance in relation to the defined objective.

Present your analysis in a concise, technically precise, and logically structured form (approximately 150-200 words). All arguments must clearly connect back to the original problem statement and use case. No new simulations or computations should be introduced.

1 Segmentation, Morphological Operations, Object Properties

Learning Objectives

- Reflective: Evaluate the accuracy, limitations, and practical applicability of segmentation results in relation to the defined use case, including sensitivity to parameters and image conditions.
- Conceptual: Apply threshold-based image segmentation to separate similar objects from the background in a domain-specific context.
- Methodological: Select and justify suitable thresholding methods, morphological operations, and object property measures based on image characteristics and the defined problem.
- Technical: Implement segmentation, morphological refinement, object extraction, and object property measurement in a structured and traceable manner.
- Analytical: Quantitatively assess segmentation performance using an appropriate measure and analyze extracted objects using simple statistical properties.
- Reflective: Evaluate the accuracy, limitations, and practical applicability of segmentation results in relation to the defined use case, including sensitivity to parameters and image conditions.

Day 11 – Data & Domain

Define the application context in which segmenting similar objects from an image is relevant. Formulate a clear use case and problem statement related to your chosen country. Explain why extracting and analyzing objects is meaningful in this context. Select one representative 2D image containing multiple similar, clearly separable objects. Objects should:

- be visually distinct from the background
- have no or minimal overlap
- be suitable for segmentation using intensity-based methods (e.g., thresholding)

Describe:

- the origin and content of the image
- the visual characteristics of the objects (e.g., brightness, contrast, shape)

Clearly define:

- what constitutes an object in your image
- the objective of segmentation (e.g., counting, measuring)

No methods or implementation should be introduced at this stage.

Start a new notebook.

Reference examples: The following use cases are provided for orientation. You may select one of them and adapt it to your chosen country, or define a suitable alternative within your selected domain.

Example 1: Agricultural Analysis – Fruit Counting and Measurement

- **Use Case:** Images from local markets or agricultural production sites are used to estimate quantities and sizes of fruits for quality control and yield assessment.
- **Problem Statement:** The image contains multiple similar objects that must be separated from the background to enable reliable counting and measurement. Variations in lighting or small artifacts may affect how clearly objects can be distinguished.

Example 2: Urban Monitoring – Vehicle Counting in Parking Areas

- **Use Case:** Images of parking areas are analyzed to estimate occupancy and support urban planning or traffic monitoring.
- **Problem Statement:** Vehicles appear as distinct objects that must be separated from the background to enable counting. Differences in illumination and shadows may make reliable separation challenging.

Day 12 – Methodological Design

Define and justify your segmentation and object analysis approach. Select a threshold-based segmentation method (e.g., global or adaptive thresholding) suitable for separating your objects from the background. Explain:

- the underlying principle
- why it is appropriate for your image and object characteristics

Define how you will refine the segmentation using morphological operations (e.g., erosion, dilation, opening, closing). Explain the intended effect of these operations on the segmentation result. Identify 1-3 key parameters (e.g., threshold value, kernel size) and explain their expected effect on the segmentation outcome. Specify how you will represent the shape of one selected object using skeletonization, and explain why this representation is meaningful for your use case.

Based on these choices, define a small number of clearly structured configurations (e.g., baseline + 1-2 variations) that enable controlled comparison. Specify:

- how objects will be extracted (e.g., connected components)
- which 2-3 object properties you will measure (e.g., area, perimeter) and why they are relevant for your use case

Limit your design to a small number of clearly structured configurations; avoid many loosely connected variations.

The focus of this day is on method selection, parameter reasoning, and structured experimental design. No implementation, results, or evaluation should be included.

Day 13 – Technical Implementation

Implement your segmentation and object analysis pipeline as defined in Day 12.

- Apply your thresholding method to generate a binary segmentation mask.
- Refine the mask using your selected morphological operations, ensuring that changes are minimal and object shapes remain largely unchanged.
- Extract the segmented objects (e.g., connected components), count them, and compute the selected 2-3 object properties.
- For the object selected in Day 12, compute the skeleton representation and report the number of skeleton pixels.

Ensure that:

- all parameter settings are explicitly stated and traceable
- the sequence of steps can be clearly followed
- key intermediate and final results are visualized

Apply the predefined configurations from Day 12 and keep the implementation focused on a small number of clearly structured configurations.

The focus of this day is on correct, transparent, and traceable implementation. No quantitative evaluation or interpretation should be included.

Day 14 – Evaluation

Quantitatively evaluate your segmentation and object extraction results.

- Select exactly one quantitative segmentation evaluation measure that is appropriate for your use case. Clearly:
  - define what the measure computes
  - justify why it is suitable for assessing segmentation performance in relation to your objective and use case
- Apply this segmentation measure consistently across all predefined configurations from Day 12.
- In addition, analyze the extracted objects by computing simple descriptive statistics (e.g., mean, standard deviation, or range) for the selected object properties.

Report:

- the segmentation metric values for each configuration
- the aggregated object property results
- a clear comparison relative to a baseline configuration

Ensure that:

- the segmentation measure is used consistently
- object properties are aggregated across all objects
- results are presented in a clear and comparable form
- differences between configurations are explicitly quantified

Limit your analysis to a small number of clearly structured configurations.

The focus of this day is on objective, quantitative assessment and structured comparison. No new simulations, theoretical derivations, or interpretative discussion should be introduced.

Day 15 – Analytical & Communication, Final Reflection & Revisions

On this day, you will observe, interpret, and discuss your findings in direct relation to your initial problem statement and defined use case. Based on the quantitative evaluation from Day 14, analyze how well your segmentation and object extraction pipeline performs and how the results translate into domain-specific consequences. Structure your analysis into:

- **Observations**: Describe the quantitative results only. Refer explicitly to the segmentation metric values and the aggregated object property results, and how they change across configurations.
- **Interpretation:** Explain why the observed changes occur based on the thresholding method, morphological operations, and parameter choices. Relate these effects to relevant image characteristics and object properties.
- **Discussion and Critical Reflection**: Assess the practical applicability of your results. Discuss how well objects are separated and measured, which configurations are suitable for the use case and why, and what limitations or risks arise (e.g., sensitivity to threshold choice, small

artifacts, incomplete separation). Reflect on how well your segmentation measure and object properties capture performance in relation to the defined objective.

Present your analysis in a concise, technically precise, and logically structured form (approximately 150-200 words). All arguments must clearly connect back to the original problem statement and use case. No new simulations or computations should be introduced.

In addition, write a personal final reflection on MC (approximately 150 words). Focus on insights and learning, not on describing what you did. Reflect on:

- what worked particularly well or impressed you and why
- what you would approach differently if you repeated the task
- which results or observations were unexpected or insightful
- what you learned about the gap between theoretical methods and practical application

Where appropriate, refer to specific decisions or outcomes from your work, but use them to support your reflection rather than to describe your process. Avoid summarizing your workflow. The focus should be on critical reflection, key takeaways, and personal learning.

Submit revisions, if applicable. Revisions are only considered if clearly highlighted in color. Code does not need to be colored; however, include a clearly marked text note in color before and after any changed code sections.